FORDHAM
THE JESUIT UNIVERSITY OF NEW YORK

Gabelli School of Business

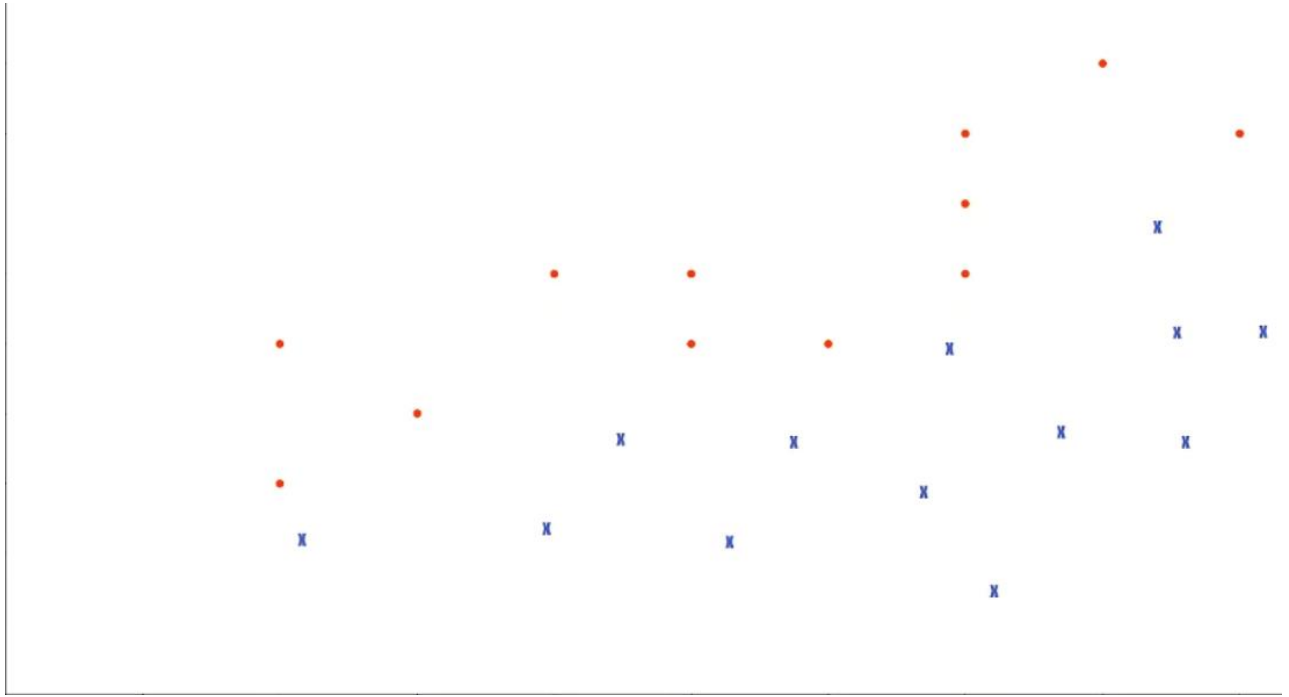
Lecture VI – Neural Networks

Machine Learning for Business

Michael Deamer

Limitation

Interaction between variables



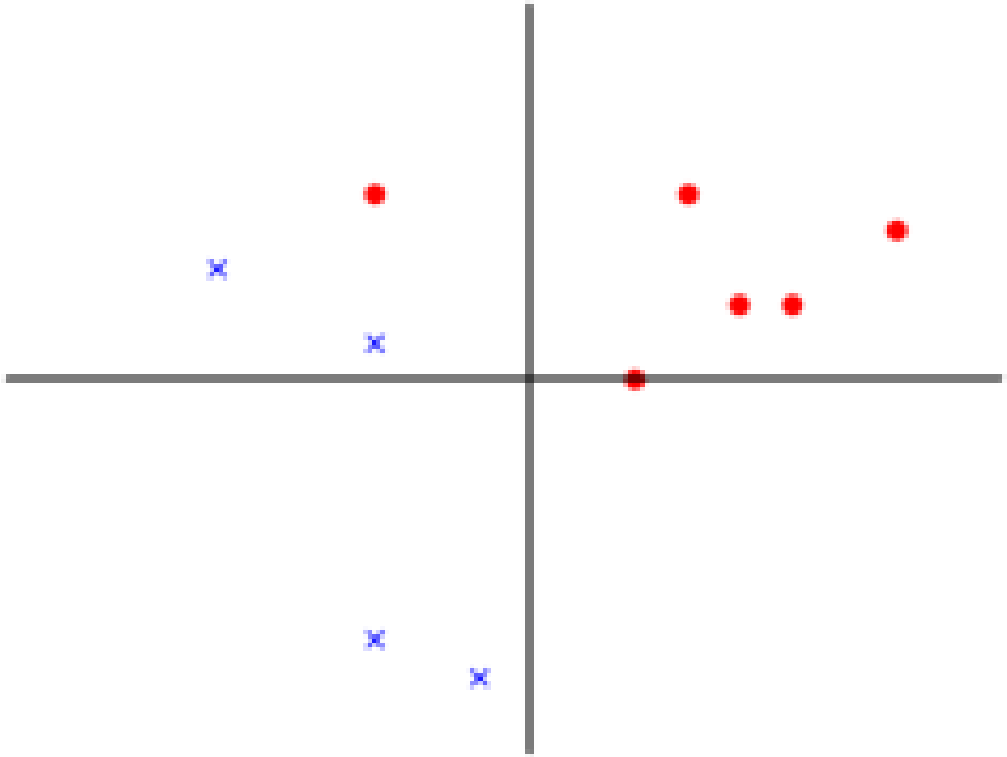
Artificial Neural Networks



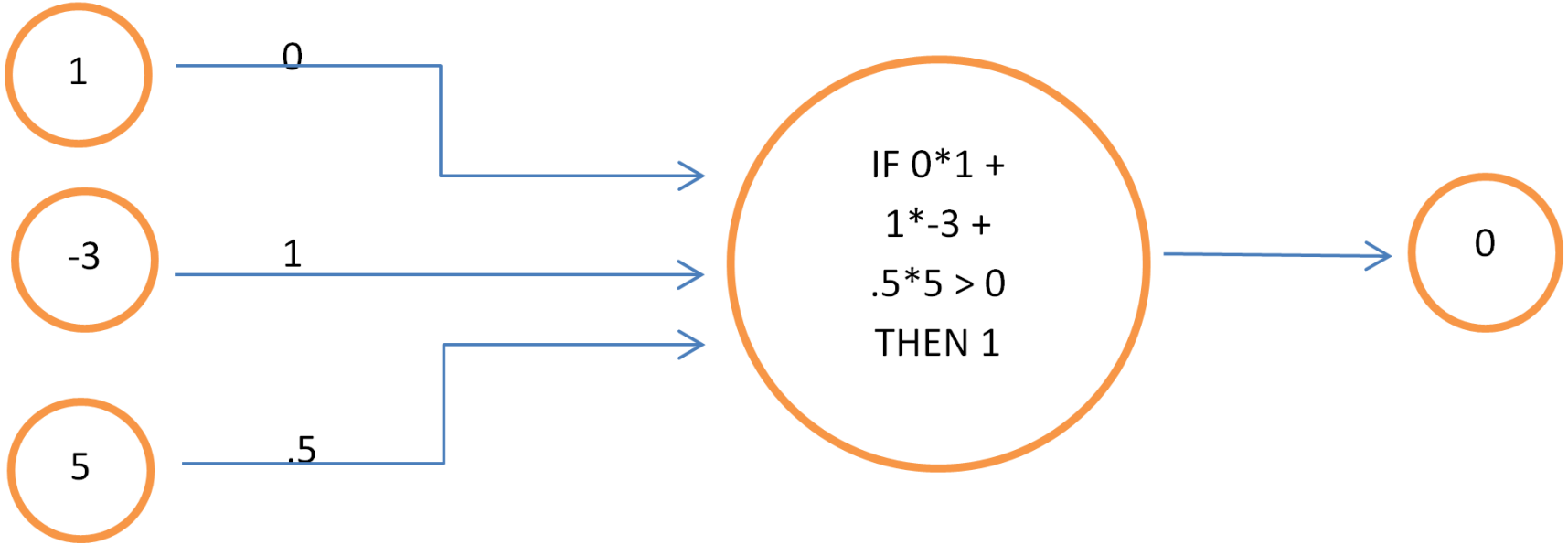
Perceptron Example

X	Y	Class
3	0	1
5	2	1
6	2	1
4	5	1
8	4	1
-3	5	1
-3	1	0
-5	3	0
-1	-9	0
-3	-8	0

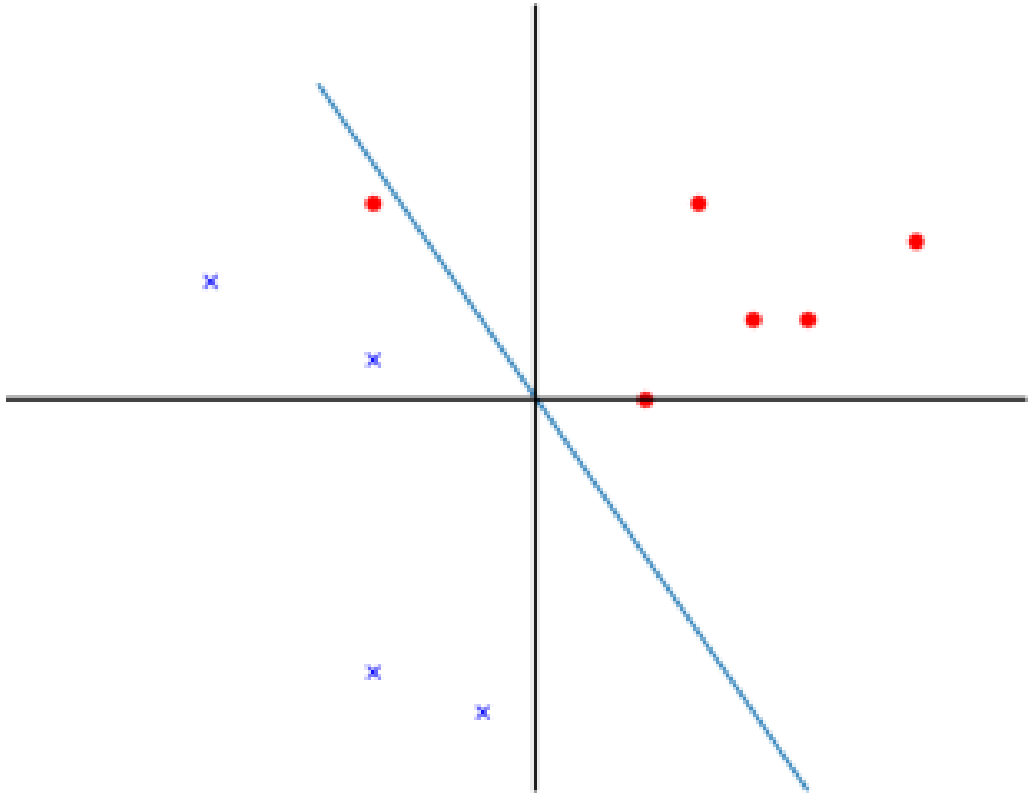
Perceptron Example



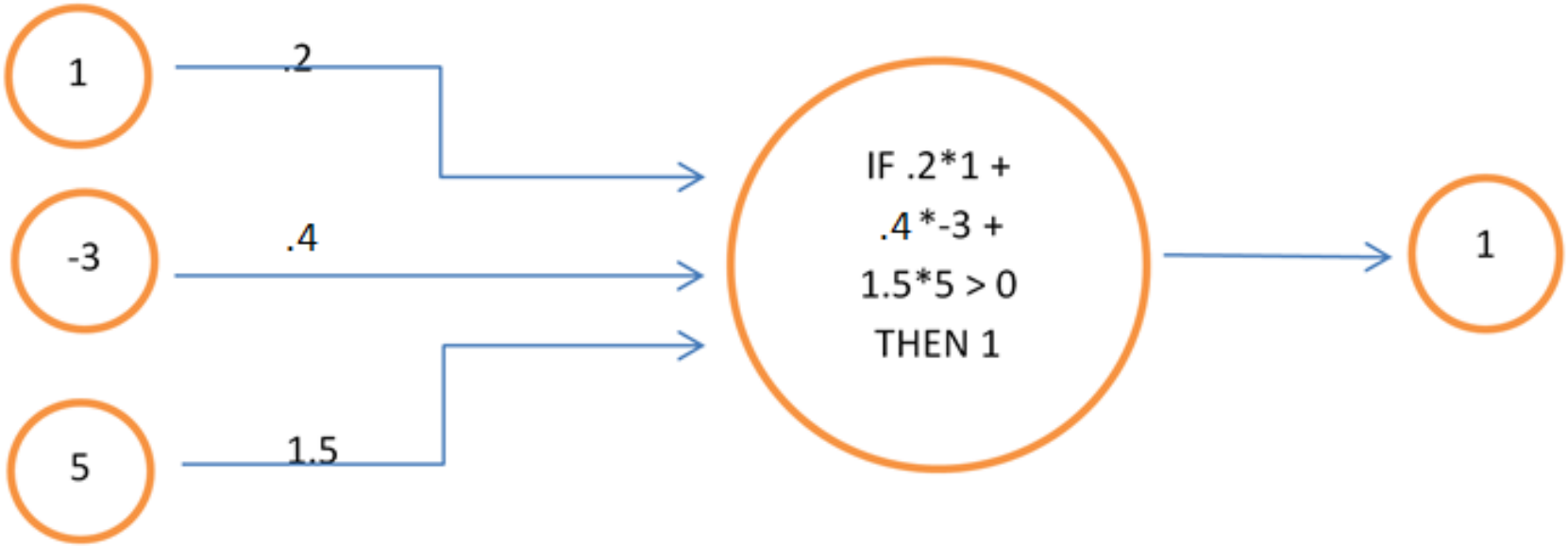
Perceptron Example



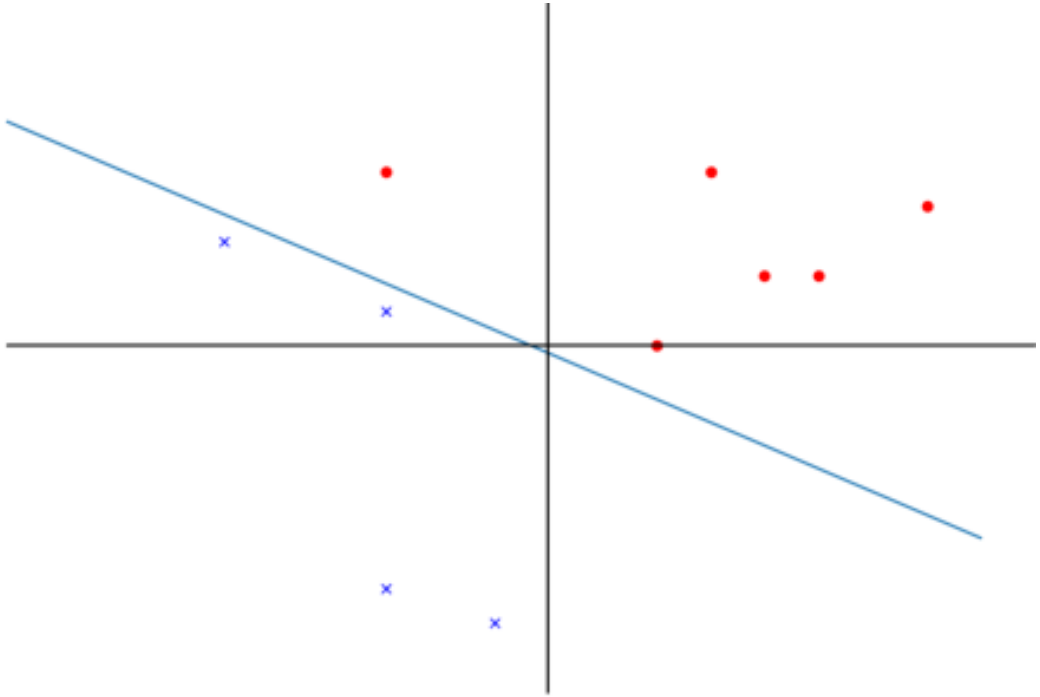
Perceptron Example



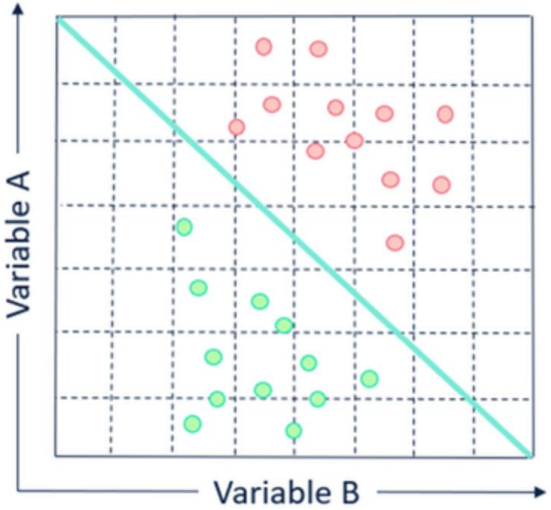
Perceptron Example



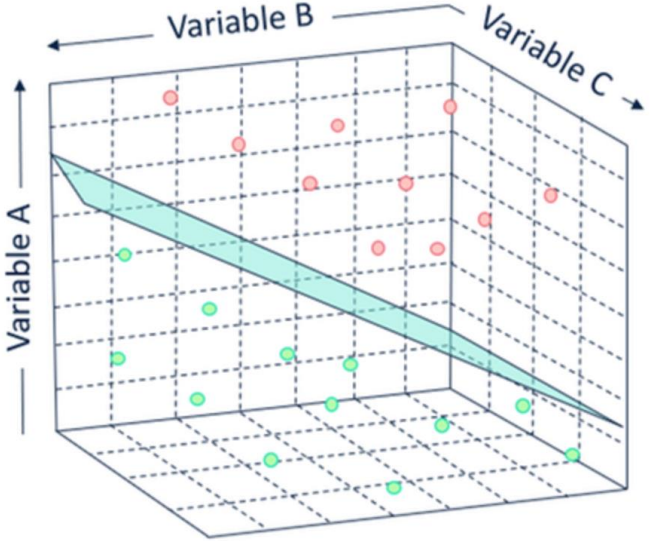
Perceptron Example



Hyperplanes



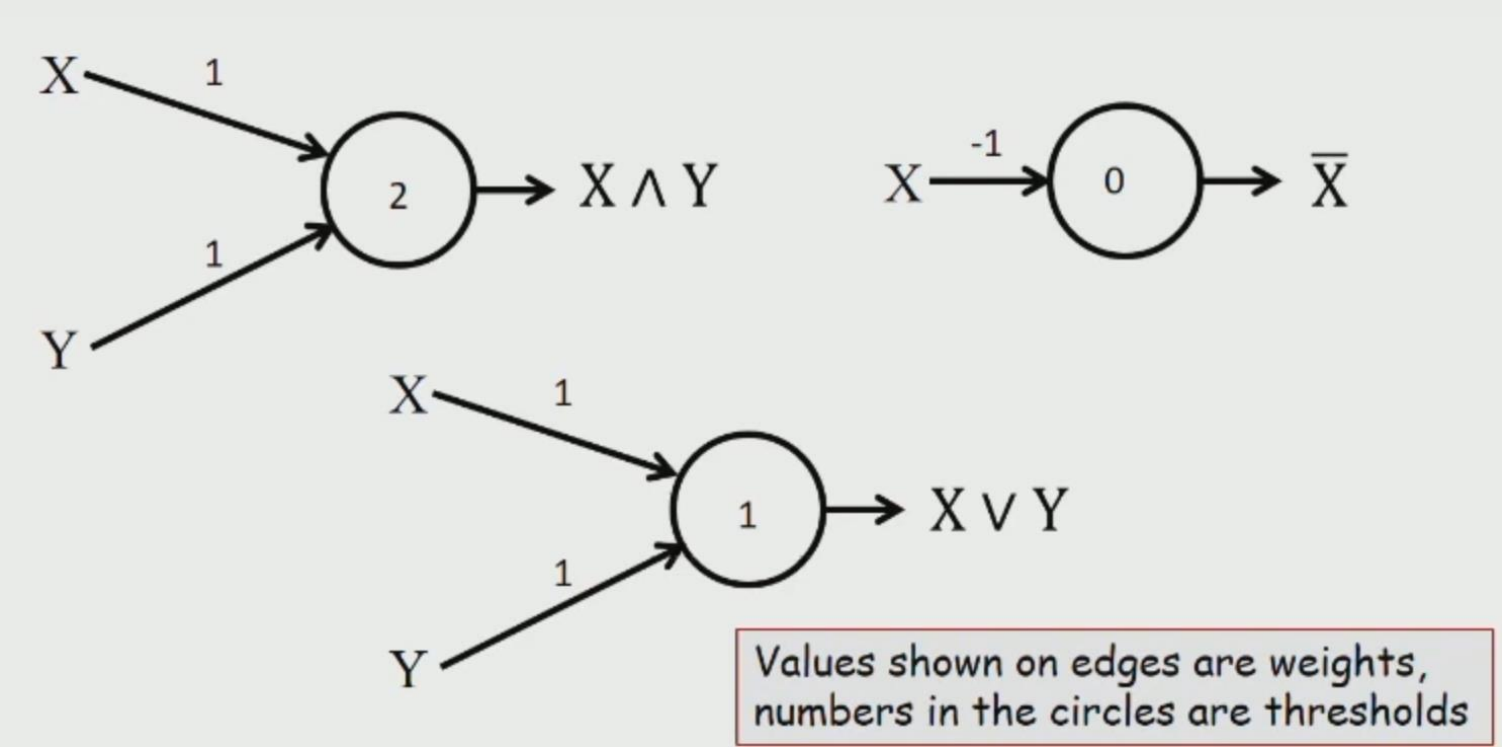
2-Dimensional Problem Space



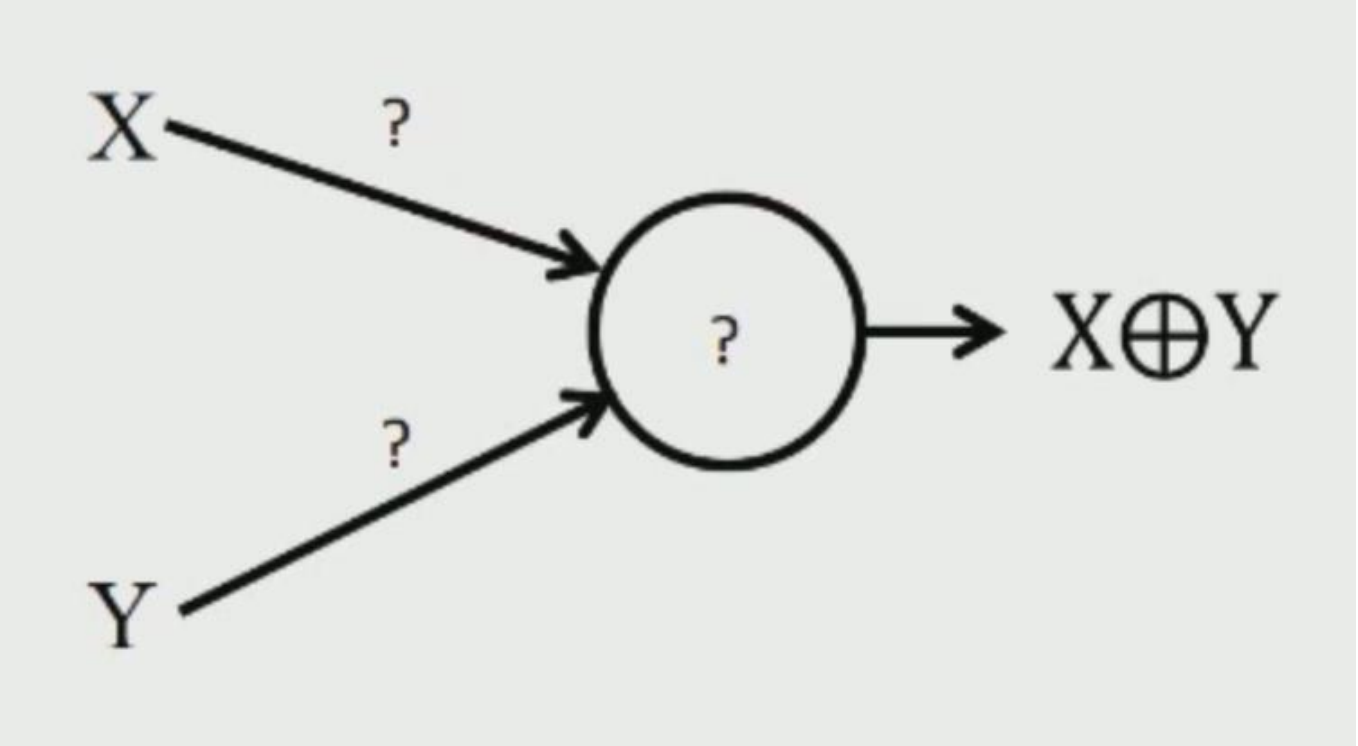
3-Dimensional Problem Space

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Go to Settings to activ

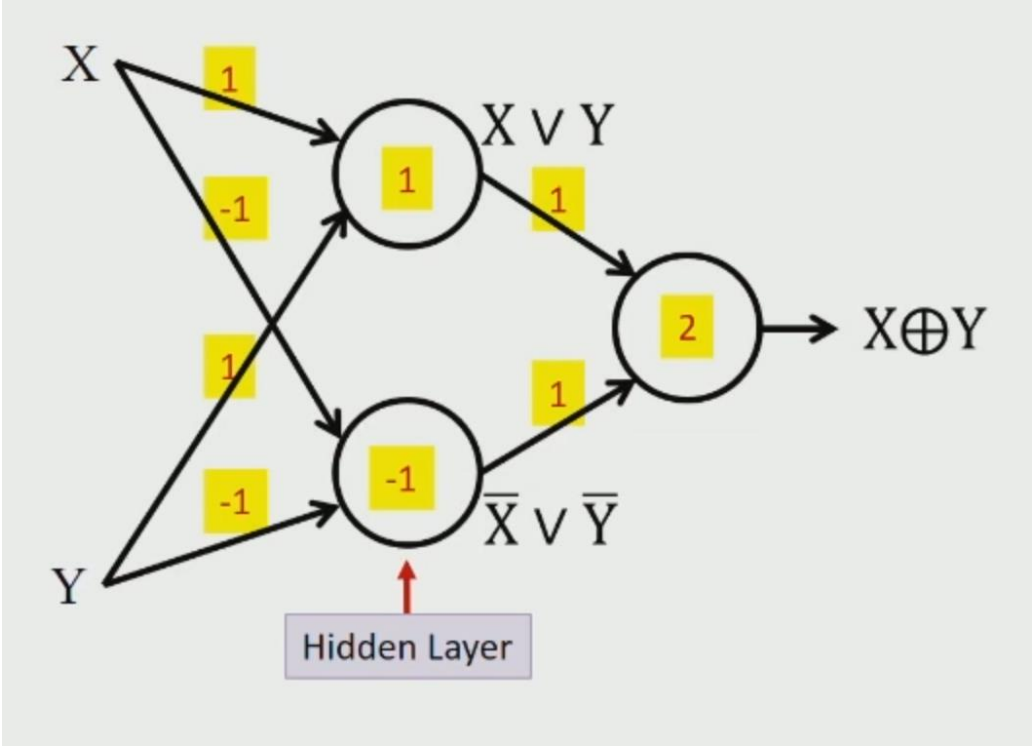
XOR


X
1
2
X ∧ Y
1
Y
X
-1
0
$\overline{X}$
X
1
1
X ∨ Y
1
Y
Values shown on edges are weights,
numbers in the circles are thresholds

XOR



XOR



To Do

- ❑ Lab V and Homework V
- ❑ Complete Project Proposal with group
- ❑ Recommended reading: 10.1

Bibliography

Tan, Steinbach, Karpatne, and Kumar. Introduction to Data Mining, second edition. Pearson, Hudson, NY, 2019.
Han, Kamber, Pei. Data Mining Concepts and Techniques, second edition. University of Illinois at Urbana-Champaign. Elsevier, San Francisco, CA, 2006.